

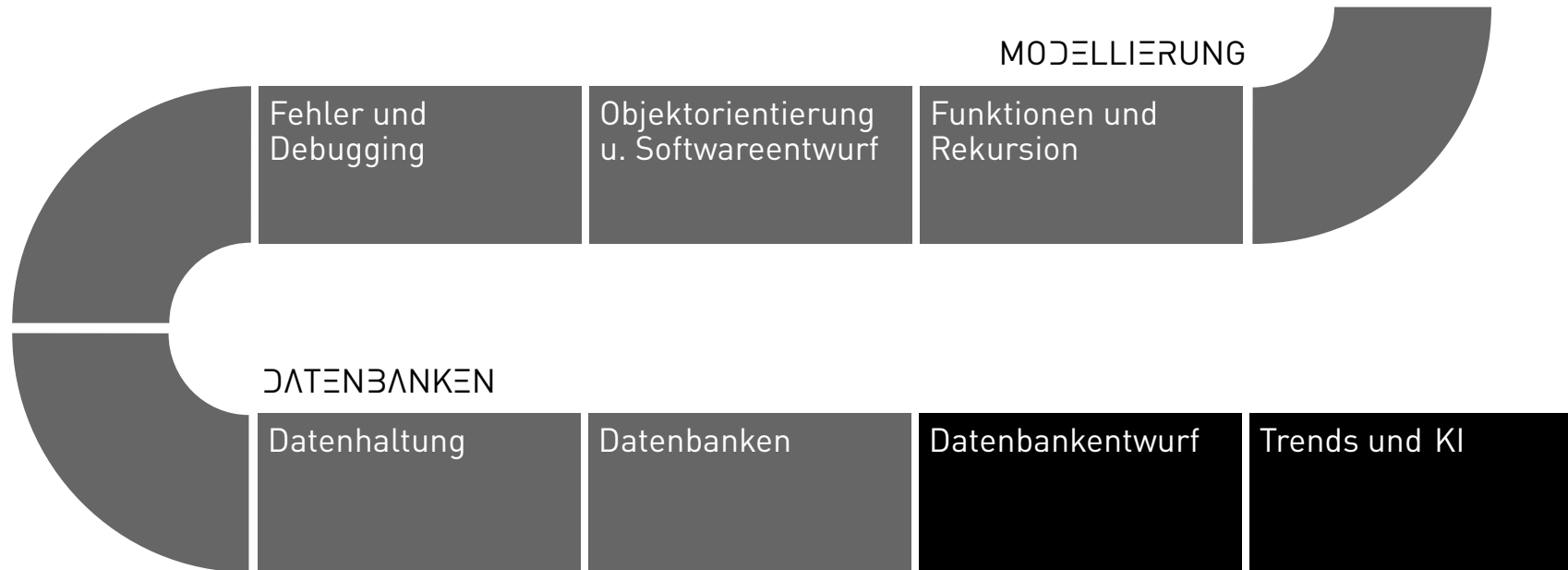
Programming and Databases

Joern Ploennigs

Relational Databases

MIDJOURNEY: TABULAR RELIEF, REF. BEN NICHOLSON

PROCESS



RELATIONAL DATABASES



Midjourney: Relational Database

RELATIONAL DATABASES - CORE CONCEPTS

Definition: Relational Database

A relational database is a *database system* that stores information in *tables* (relations), where the data are organized as *tuples* (rows) and each column (attribute) has a defined data type.

- Data are organized in table structures (schemas)
- *Relation* is the term for a table
- A *relational database* is a set of tables

Column 1 Column 2 Column 3

Row 1	...	...
Row 2	...	...

RELATIONAL DATABASES - BASIC CONCEPTS

- Jede Tabelle hat einen *relation name* (table name)
- *Relation* denotes the set of all data tuples (all rows)
- Jedes *tuple* an Daten besteht aus einer Abfolge von Attributwerten (Einzelne Zeile)
- Reihenfolge festgelegt durch *relation schema* (Spaltennamen)

Attribute 1	Attribute 2	Attribute 3
Attribute value ...		...
Attribute value ...		...

EXAMPLE: OPENDATA HANSEATIC CITY OF ROSTOCK

Information is usually spread across several tables and often needs to be merged and processed to derive actionable knowledge.

- Example: <https://www.opendata-hro.de>
 - Land-use plans
 - Municipalities
 - Construction sites
 - Standard land values
 - Address list

KEY ATTRIBUTES

- Certain *attributes (columns)* can be marked as *keys*
- Keys are usually *uniquely identifiable*
- They exist in the table only *once*
- They can be used to identify the *row (tuple)*
- Row numbers are not suitable (they can be deleted)

Key types:

- A *single attribute* or a *set of attributes*
- *Unique ID (UID)* - a single attribute
- *Globally Unique IDs (GUID)* - globally unique
- *UUID* (universally unique identifier) - ISO standard

KEY ATTRIBUTES - EXAMPLE

Municipalities Table

MunicipalityID	Name	Population
1	Dummerstorf	7.329
2	Graal-Müritz	4.278
3	Sanitz	5.831

Key: MunicipalityID

KEY REFERENCES

Municipalities Table

MunicipalityID	Name	Population
1	Dummerstorf	7,329
2	Graal-Müritz	4,278
3	Sanitz	5,831

Primary Key: MunicipalityID

Buildings Table

BuildingID	BuildingType	MunicipalityID
5000	Gas Station	2
5001	Hotel	1
5002	Church	2

Foreign Key: MunicipalityID

KEY CONCEPTS

- The *keys of other tables* can be used to refer to rows in those tables
- A table's own keys are called *Primary Key* (Primary Key)
- The *keys of other tables* are called *Foreign Key* (Foreign Key)
- The *tables are related through keys*

The foreign key references the primary key in the other table

THE ROLE OF KEYS

Keys define the so-called integrity constraints:

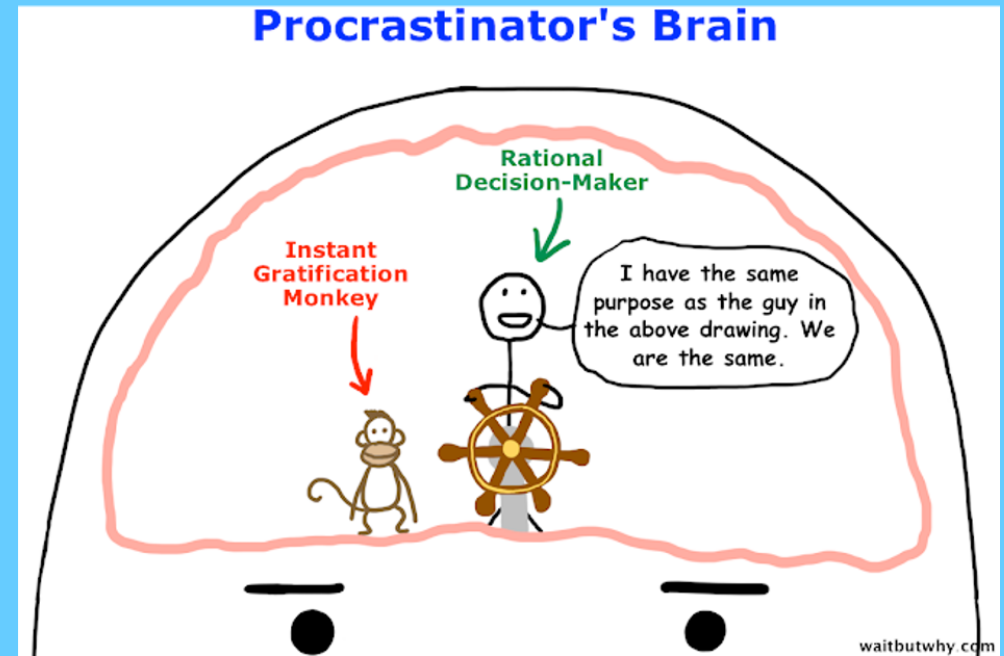
Primary keys

- Represent the *integrity of individual tables*
- *Local integrity*

Foreign keys

- Represent the *integrity of the entire database system*
- *Global integrity*

LESSON LEARNED



<https://youtu.be/arj7oStGLkU>

LESSONS LEARNED - PROKRASTINATION

- Some procrastination is normal and often part of working life (it follows from prioritization)
- Procrastination is very dangerous when we misestimate the effort and it turns out to be larger than the available time

LESSONS LEARNED - TIME MANAGEMENT

Questions?

